

TIC 59490510

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 9.124822 d
Epoch (t0) 2503.5152 BTJD
Duration 0.1200 d
Depth 32.990 % (329,905 ppm)
SNR 1068.4

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

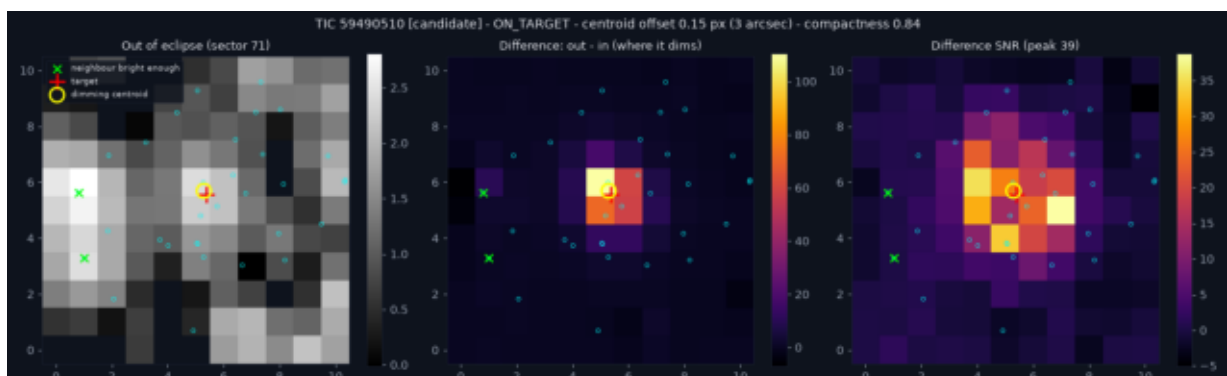
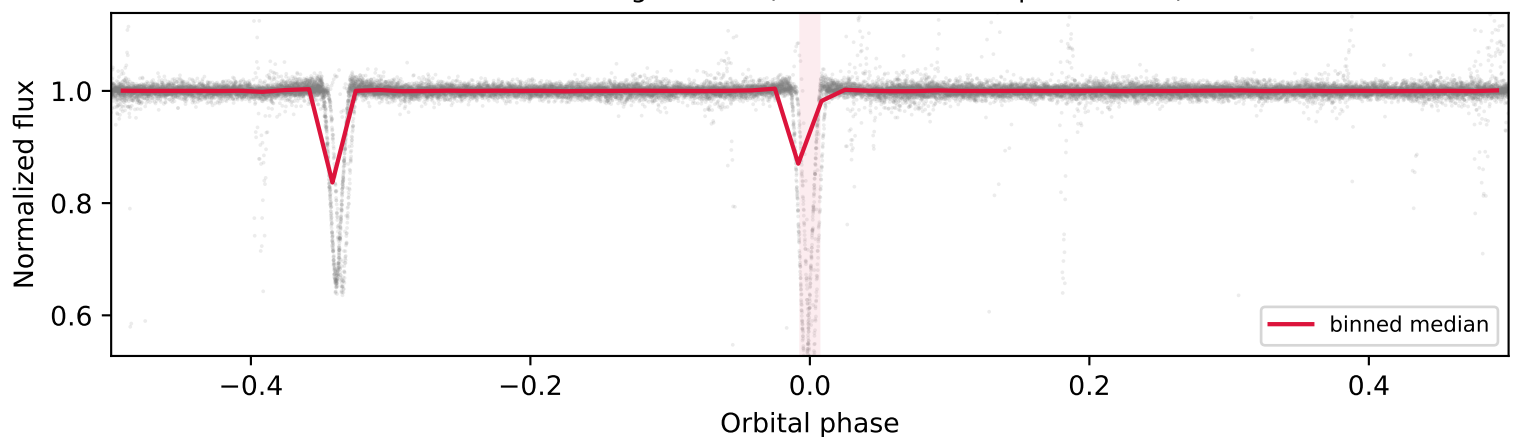
Eclipses observed 12 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 761.9 (gate ≥ 7.0)
Odd/even z-score 2.65 (consistent)
Secondary eclipse detected, phase -0.33, 9.7 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 71 light-curve dip in this sector: 310.7 sigma t0 refined by -0.0050 phase
Difference-image peak SNR 38.74 (gate ≥ 5.0) Compactness 0.84 (gate ≥ 0.50)
Dimming centroid offset 0.15 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 59490529
4.49 px away, dTmag -0.88 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.